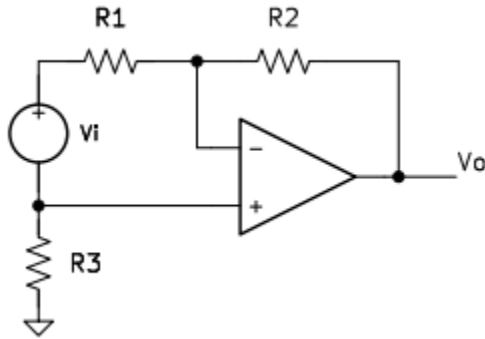




Using the circuit as drawn above, which is driven by an ideal differential voltage source, do the following steps:

(i) Find the feedback factor, b , of the system. Reminder: to do this, you assume ideal conditions.

$$v_+ = R_3(V_{in} - V_{out}) / (R_1 + R_2 + R_3)$$

$$v_- = (R_1 + R_3)(V_{in} - V_{out}) / (R_1 + R_2 + R_3) + V_{in}$$

$$b = (v_+ - v_-) / V_{out} \text{ (set } V_{in} = 0 \text{)}$$

$$b = (R_3(-V_{out}) / (R_1 + R_2 + R_3) - (R_1 + R_3)(-V_{out}) / (R_1 + R_2 + R_3)) / V_{out} = (R_1) / (R_1 + R_2 + R_3)$$

$$b = (R_1) / (R_1 + R_2 + R_3)$$

(ii) Solve for the closed loop gain using exact analysis while incorporating the effects of r_d , r_o , and a finite a . Do not make assumptions on the relative sizes of these values to R_1 , R_2 , and R_3 . You do not need to significantly simplify your answer. Once you have $A = V_o/V_i$ in terms using only the circuit parameters, you may stop.

$$V_{out} = a(v_+ - v_-) - i_o r_o$$

$$(v_+ - 0)/R_3 + (v_+ - v_-)/r_d + (v_+ - v_-)/R_1 = 0$$

$$v_+/R_3 + v_+/r_d + -v_-/r_d + v_+/R_1 + -v_-/R_1 = 0$$

$$(v_- - V_{out})/R_2 + (v_- - v_+)/r_d + (v_- - v_+)/R_1 = 0$$

$$v_-/R_2 - V_{out}/R_2 + v_-/r_d - v_+/r_d + v_-/R_1 - v_+/R_1 - V_{in}/R_1 = 0$$

$$i_o = (V_{out} - v_-)/R_2$$

$$V_{out} = a(v_+ - v_-) - r_o(V_{out} - v_-)/R_2$$

$$V_{out} = a(v_+ - v_-) + (r_o v_-)/R_2 - V_{out}(r_o)/R_2$$

$$V_{out}(1 + (r_o)/R_2) = a(v_+ - v_-) + (r_o v_-)/R_2$$

(iii) Solve for the closed loop gain using return ratio analysis. You do not need to significantly simplify your answer. Once you have T and a_f in terms using only the circuit parameters, you may stop.

(iv) Note that the above two steps should produce the exact same equations, although they may not be formatted the same way symbolically. Check your work by simplifying each with $r_{dd} = \infty$ and $r_o = 0 \Omega$. If correct, you should produce the below gain for each.

$$A = - \left(\frac{R_2 + R_3}{R_1} \right) / \left(1 + \frac{(R_1 + R_2 + R_3)}{a R_1} \right) = - \frac{(R_2 + R_3)}{R_1 + (R_1 + R_2 + R_3)/a}$$

(v) Using Blackman's Formula, find the input impedance, R_i , the output impedance, R_o , and the circuit impedance as seen by R_2 , R_{c2} . For the both R_i and R_{c2} , the "nodes" in the analysis would be each side of the voltage source or resistor, respectively. For ease of analysis, you may assume $r_d \gg R_1, R_2, R_3 \gg r_o$.

P2) Book Problem 1.61. Note: P1.15-P1.19 refers to the figures associated with problems 1.15-1.19, not the chapter figures 1.15-1.19

1.4 Ideal op amp circuit analysis

- 1.15 Find v_N , v_P , and v_O in the circuit of Fig. P1.15, as well as the power released by the 4-V source; devise a method to check your results.

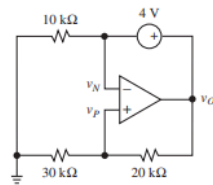


FIGURE P1.15

- 1.16 (a) Find v_N , v_P , and v_O in the circuit of Fig. P1.16. (b) Repeat (a) with a 5-kΩ resistance connected between A and B.

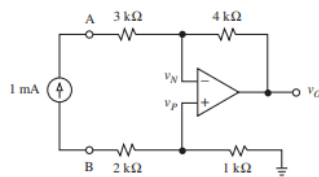


FIGURE P1.16

- 1.17 (a) Find v_N , v_P , and v_O in the circuit of Fig. P1.17 if $v_S = 9$ V. (b) Find the resistance R that, if connected between the inverting-input pin of the op amp and ground, causes v_O to double. Verify with PSpice.

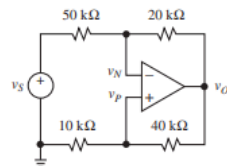


FIGURE P1.17

- 1.18 (a) Find v_N , v_P , and v_O in the circuit of Fig. P1.18. (b) Repeat (a) with a 40-kΩ resistance in parallel with the 0.3-mA source.

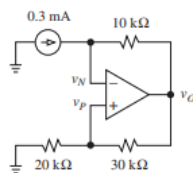


FIGURE P1.18

- 1.19 (a) Find v_N , v_P , and v_O in the circuit of Fig. P1.19 if $i_S = 1$ mA. (b) Find a resistance R that when connected in parallel with the 1-mA source will cause v_O to drop to half the value found in (a).

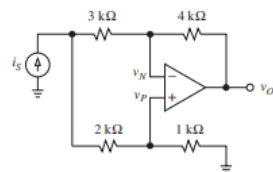


FIGURE P1.19

I believe all these sources need to be treated as the input? doesn't really work otherwise?

$$V_{out} = 10^3 (v_+ - v_-) = 10^3 V_{test}$$

15:

$$v_+ = V_{out} (30k/(30k+20k)) = 0.8 V_{out} = 0.8 * 10^3 * V_{test}$$

$$v_- = V_{out} - V_{in} = 10^3 * V_{test} - V_{in} \text{ (set } V_{in}=0) = 10^3 V_{test}$$

$$T = - (v_+ - v_-)/V_{test} = -(0.8 * 10^3 V_{test} - 10^3 V_{test})/V_{test} = 0.2 * 10^3 = 200$$

16:

$$v_+ = -\text{lin} * 1k \text{ (lin} = 0) = 0$$

$$v_- = V_{out} + \text{lin} * 4k \text{ (lin} = 0) = V_{out} = 10^3 V_{test}$$

$$T = - (v_+ - v_-)/V_{test} = - (0 - 10^3 V_{test})/V_{test} = 10^3 = 1000$$

17:

$$v_+ = V_{out} 10k/(10k+40k) = .2 V_{out} = .2 * 10^3 V_{test}$$

$$v_- = (V_{out} - V_{in}) 50k/(50k+20k) \text{ (} V_{in}=0) = .714 V_{out} = .714 * 10^3 V_{test}$$

$$T = - (v_+ - v_-)/V_{test} = - (.2 - .714) * 10^3 = 514.286$$

18:

$$v_+ = V_{out} 20k/(20k+30k) = .4 * 10^3 V_{test}$$

$$v_- = V_{out} + \text{lin} * 10k \text{ (lin} = 0) = V_{out} = 10^3 V_{test}$$

$$T = - (v_+ - v_-)/V_{test} = - (.4 - 1) * 10^3 = 600$$

19:

$$v_+ = \text{lin} * (3k+4k)/(3k+4k+2k+1k) * 1k = \text{lin} * 700 \text{ (lin} = 0) = 0$$

$$v_- = \text{lin} * (2k+1k)/(3k+4k+2k+1k) * 4k + V_{out} = \text{lin} * 1200 + V_{out} \text{ (lin} = 0) = 10^3 V_{test}$$

$$T = - (v_+ - v_-)/V_{test} = - (-10^3 V_{test})/V_{test} = 10^3 = 1000$$

P3) ECSE-6050 ONLY: Book Problem 1.64.

1.64 (a) Assuming the op amp of Fig. P1.64 has $r_d \cong \infty$, $a = 10^3$ V/V, and $r_o \cong 0$, and all resistances are identical, find A_{ideal} as well as the gain error GE. (b) Find a_{min} for $GE \leq 0.1\%$.

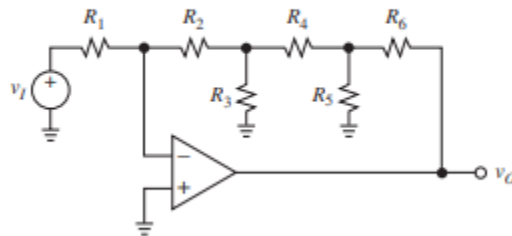


FIGURE P1.64

a) Aideal, GE

$$v_+ = 0$$

$$v_- = \frac{8}{13} v_{in} + \frac{1}{13} v_{out}$$

$$v_+ = v_- ; 0 = \frac{8}{13} v_{in} + \frac{1}{13} v_{out} ; v_{out} = -8 v_{in} ; A_{ideal} = -8$$

$$v_{out} = 10^3 (v_+ - v_-)$$

$$v_{out} = 10^3 \left(-\frac{8}{13} v_{in} - \frac{1}{13} v_{out} \right) = -\frac{8}{13} 10^3 v_{in} - \frac{1}{13} 10^3 v_{out}$$

$$v_{out} \left(1 + \frac{1}{13} 10^3 \right) = -\frac{8}{13} 10^3 v_{in}$$

$$v_{out}/v_{in} = -\frac{8}{13} 10^3 / \left(1 + \frac{1}{13} 10^3 \right) = -7.897 = A_{real}$$

$$GE = (A_{ideal} - A_{real}) / A_{ideal} = 0.0128$$

b) set $GE=0.001$,

$$(-8 - A_{real}) / -8 = 0.001, A_{real} = -7.992$$

$$-7.992 = -\frac{8}{13} a / \left(1 + \frac{1}{13} a \right)$$

$$a_{min} = 12987$$

P4) Book Problem 1.74.

1.74 (a) Assuming $I_Q = 50 \mu\text{A}$ and a grounded load of $100 \text{ k}\Omega$ at the output of the dc-offsetting amplifier of Fig. 1.17, find the values of v_I for which the op amp dissipates the maximum power. Show all corresponding voltages and currents. (b) Assuming

$\pm V_{\text{sat}} = \pm 12 \text{ V}$, find the range of values of v_I for which the op amp still operates within the linear region.

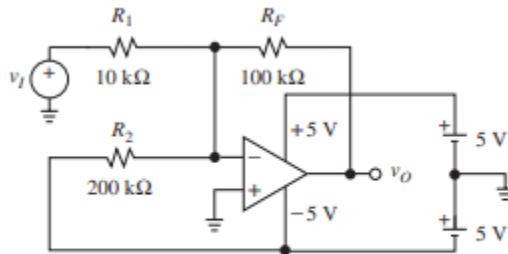


FIGURE 1.17

A dc-offsetting amplifier.

a) max PD

$$-V_{in}/10\text{k} + -(-5)/200\text{k} + -V_{out}/100\text{k} = 0$$

$$-10V_{in} + 2.5 - V_{out} = 0$$

$$V_{out} = -10V_{in} + 2.5$$

$$i_{out} = i_L + i_{R3} = V_{out}/100\text{k} + V_{out}/100\text{k} = 2V_{out}/100\text{k}$$

$$PD = P_{supplies} - P_{delivered} = (5+5)I_Q = 10 \cdot 50\mu\text{A} = 0.5\text{mW}$$

$$P_{delivered} = P_L = V_{out}^2/RL = V_{out}^2/100\text{k}$$

dissipation is largest when load power is smallest, $V_{out}=0$

$$0 = -10V_{in} + 2.5$$

$$V_{in} = 0.25\text{V}$$

$$PD = 0.5\text{mW}$$

b) linear region

$$-12 < V_{out} < 12$$

$$-10V_{in} + 2.5 = V_{out}$$

$$-10V_{in} = V_{out} - 2.5$$

$$V_{in} = 0.25 - V_{out}/10$$

$$-12: V_{in} = 0.25 + 1.2 = 1.45$$

$$12: V_{in} = 0.25 - 1.2 = -.95$$

$$-.95 < V_{in} < 1.45$$

P5) Book Problem 1.77. Note: Fig. P1.17, NOT Fig. 1.17. Instead of using the input $\pm 9\text{V}$ sine wave as requested by the book, use a $\pm 25\text{V}$ triangular wave.

1.77 Assuming the op amp of Fig. P1.17 saturates at $\pm 10\text{ V}$, sketch and label v_N , v_P , and v_O versus time if v_S is a sinusoidal wave of $\pm 9\text{-V}$ peak values.

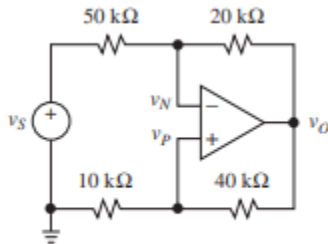


FIGURE P1.17

$$v_+ = V_{out} \cdot 0.2$$

$$v_- = (V_{in} - V_{out}) \cdot \frac{5}{7} + V_{in}$$

$$V_{out} \cdot 0.2 = (V_{in} - V_{out}) \cdot \frac{5}{7} + V_{in}$$

$$V_{out} = V_{in} \cdot \frac{12}{7} / (0.2 + \frac{5}{7}) = 1.875 V_{in}$$

$$V_{out} = \max(-10, \min(10, 1.875 V_{in}))$$

$$v_+ = V_{out} \cdot 0.2$$

$$v_- = (V_{in} - V_{out}) \cdot \frac{5}{7} + V_{in}$$

